

Eway Bill Document Parser

Converts Eway Bill PDFs → Structured JSON automatically

Prepared by

Adarsh

Date

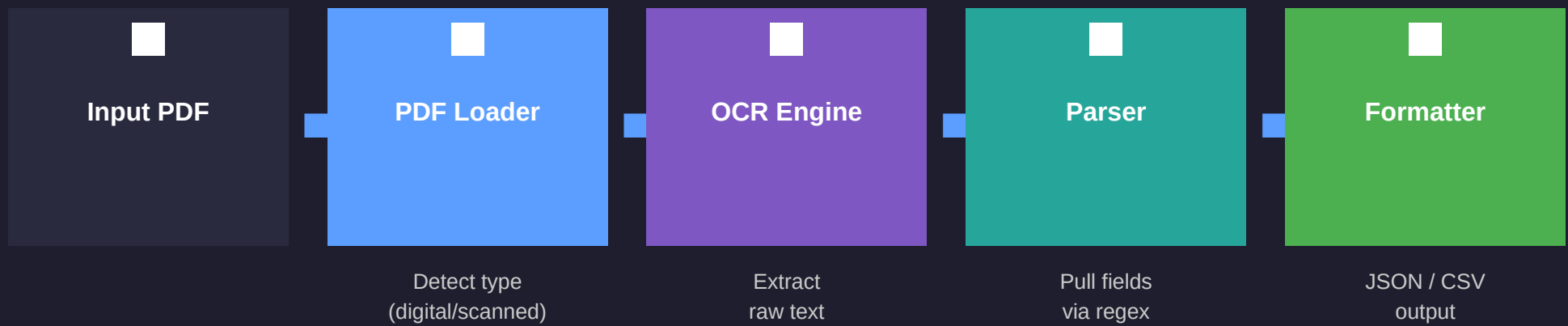
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Status

Draft — Pending Review

What are we building?

- 1 Accept an Eway Bill PDF as input
- 2 Automatically detect if PDF is digital or scanned
- 3 Extract all relevant fields using OCR
- 4 Return structured data in JSON format
- 5 Plug into any downstream application — fully independent module



Module 1 — PDF Loader

Load the PDF from file path

Attempt text extraction with pdf-parse

Text > 50 chars → Digital PDF

Text empty / minimal → Scanned PDF

Output fields:

path, filename, numPages,
isDigital, rawBuffer, pagesText[]

Module 2 — OCR Engine

Routes based on PDF type:

Digital PDF:

→ Return text directly (fast)

Scanned PDF:

→ Convert pages to images

→ Run tesseract.js (WASM-based)

→ Return extracted text

Fields extracted from every Eway Bill

EWB Details	From Party	To Party	Item Details	Transporter
EWB Number	GSTIN	GSTIN	HSN Code	Transporter GSTIN
EWB Date	Company Name	Company Name	Product Desc.	Name
Valid Until	Place	Place		Vehicle No.
Txn Type	State	State		Mode
Doc No & Date	Pincode	Pincode		
Value of Goods				
IRN				

Module 4 — Formatter

Saves parsed data to disk

JSON (primary)

→ Pretty-printed nested structure

→ Used by downstream apps

CSV (secondary)

→ Flattened single-row format

→ For manual review / Excel

Naming: {filename}_{timestamp}.json

Module 5 — Main Entry Point

Wires all 4 modules together

Accepts CLI arguments

Commands:

`ts-node main.ts <pdf>`

→ outputs JSON

`ts-node main.ts <pdf> --csv`

→ also outputs CSV

`ts-node main.ts <pdf> --raw`

→ prints raw text

TypeScript	Language	Type safety, better maintainability
Node.js	Runtime	Cross-platform, no server needed
pdf-parse	Digital PDF Extraction	Pure JS — no system dependencies
tesseract.js	Scanned PDF OCR	WASM-based — no system install needed
JSON + CSV	Output Formats	JSON for apps, CSV for manual review

Why no cloud API? → Free, local, no rate limits, sustainable at any volume

Phase 1	PDF Loader + Digital text extraction	DONE
Phase 2	Parser — all Eway Bill fields via regex	DONE
Phase 3	JSON + CSV formatter	DONE
Phase 4	Scanned PDF support via tesseract.js	PENDING
Phase 5	Batch processing — folder of PDFs	PENDING
Phase 6	REST API wrapper (optional)	PENDING

Seeking answers before Phase 4 implementation begins

Q1

Multi-item Eway Bills

Should the module handle documents with multiple HSN codes / items per bill?

Q2

API vs CLI

Do we need a REST API wrapper, or is CLI usage sufficient for now?

Q3

JSON Schema

Should output JSON follow a specific schema already used in the system?

Q4

Future Document Types

Are there more docs to handle — invoices, delivery challans, etc.?

Ready for Review

Phases 1–3 are complete and working on real Eway Bill PDFs

5 Modules

Independent, plug-and-play

2 Output Formats

JSON + CSV

0 Cloud APIs

Fully local, sustainable

3 Phases Done

Working on real PDFs today